Credit Risk Modeling

Project Overview

Credit Risk Modeling is a **data analytics and machine learning project** designed to predict the creditworthiness of customers. The model analyzes financial and behavioral data to determine the probability of loan default.

Problem Statement

Financial institutions face significant risk when lending money. Predicting whether a customer will default on a loan helps minimize financial losses.

Objectives

- Develop a predictive model for credit risk assessment.
- Identify high-risk customers.
- Assist lenders in making informed decisions.

Technologies Used

Programming Language - Python
Data Analysis - Pandas, NumPy
Machine Learning - Logistic Regression, Decision Trees, Random Forest
Model Evaluation - Scikit-learn, ROC-AUC Score, Confusion Matrix
Data Visualization - Matplotlib, Seaborn
Business Intelligence - Power BI
Database - SQL
Tools & Environment - Jupyter Notebook, GitHub

Dataset

Financial datasets containing:

- Customer income
- Credit history
- Loan amount
- Repayment behavior

Methodology

1. Data cleaning and preprocessing
2. Exploratory Data Analysis (EDA)
3. Feature selection
4. Model training using Logistic Regression
5. Performance evaluation

Key Features

- Predictive credit scoring
- Risk segmentation
- Data visualization dashboards

Results

The model successfully identifies high-risk borrowers and improves loan approval decisions.